MILITARY SPECIFICATION

PROJECTOR, RETICLE IMAGE: 10547353

1. SCOPE

1.1 This specification covers a sub-assembly of the Sight, Reflex: XM70E1, known as Projector Reticle Image: 10547353, referred to herein as the projector (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-P-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-81O	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester - Two Station
A8658940	Inspection Aid (Gunner's Quadrant)
F10547353	Projector Reticle Image: 10547353

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F1O552O1O	Gage fixture optical
F10552O37	Gage fixture optical

PACKAGING DATA SHEETS

10547353	Packaging of Projector, Reticle Image: 10547353

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication - The projector shall be manufactured in accordance with Drawing F10547353, Projector, Reticle Image and drawings pertaining thereto.

3.3 General Specifications - The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830

(a) Lenses
(b) Cleanliness

3.3.2 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Storage at extreme temperatures - The projector shall not evidence damage after storage at -65° Fahrenheit (F) and +160°F.

3.5 Operability at extreme temperature - Requirements 3.5.1 and 3.5.2 shall be met while the projector is thermally stabilized at -40°F and at +150°F.

3.5.1 Light Intensity Control and Light Filament Selector Switch - The light intensity control and light filament selector switch shall be manually operable.

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3.5.2 Ann 10547473 - The torque values exhibited by Arm 10547473 shall not exceed the values demonstrated for each position specified in 3.7.10 by more than 10 pound inches.

3.6 Humidity - The projector shall not evidence damage after exposure to a 97% relative humidity (R H) for 24 hours at +160"F followed by four hours at between +60"F and +90"F with condensation taking place on the projector.

3.7 Performance between +60°F and +90"F - Except where otherwise specified, the projector shall meet the requirements of 3.7.1 to 3.7.8 after orientations "a" to 'Id" inclusive have been effected. This orientation shall establish a geometric axis designated as line "w" in relation to all other axes depicted in Figure 1 and as defined in the following requirements:

a. In a plane that is 4.5 inches above and parallel to the horizontal longitudinal plane containing line A (Figure 1). Line A shall be an axis through the center of the bore in the cast-in leg, or support, provided on the main housing to mount the projector to its next higher assembly.

b. In a plane 90° to a plumb plane containing Line D (Figure 1). Line D shall be an axis through the center of the bore in the cast-in yoke on the main housing which supports the tapered arm assembly.

c. In a plumb longitudinal plane that is parallel and 17-5/16 inches from a vertical longitudinal plane containing Line D (Figure 1) in the direction of the beamsplitter-when Arm 10547473 is in an operating position.

d. In a plumb longitudinal plane that forms an angle of 40° i 2° with a plumb plane containing the center line of datum diameter C on Drawing F10547482.

3.7.1 Vibration - The projector shall evidence no damage or electrical circuit malfunction during or after the vibration specified herein, with electrical potential applied, and Arm 10547473 in an operating and stowed position. The initial optical-mechanical alignment demonstrated while meeting 3.7.2 shall not have changed more than one mil subsequent to either vibration. There shall be no flickering'of the projected reticle image during vibration.

3.7.1.1 Vibration "A" - This vibration shall be for two hours, in each plane of three mutually perpendicular planes, at the sinusoidal frequencies and amplitudes depicted in Figure 2. During each first hour period, the frequency shall be cycled logarithmically from 5 to 500 to 5 cycles per second (cps) for four cycles each of 15 t 1 minute duration. A fifteen minute dwell shall occur at each of the most significant resonant frequencies (not exceeding 4) exhibited by the sight during the first hour of each two hour vibration in each plane. Arm I05L7473 shall be sequentially located in a stowed and operating position when meeting

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this requirement, and Reticle Lamp 10547031 shall not be subjected to the vibration dwells specified herein.

3.7.1.2 Vibration B - The projector's projected reticle image shall not be displaced than ± 5 roils during this vibration from its initial position prior to vibration. This vibration shall be in a vertical plane for 5 minutes ± 15 seconds at a frequency of 32 cps with an amplitude of 0.015 inch (0.03 inch total excursion).

3.7.2 Optical - Mechanical Alignment - The center of the projector's reticle image shall be coincident to line W within 1.5 roils when *viewed* **through the central area of the beamsplitter.**

3.7.3 Optical-Mechanical Alignment Retention - The optical-mechanical alignment demonstrated by the projector in 3.7.2 shall not change more than ± 0.5 mil when Arm 10547473 is repeatedly moved from its operating to stowed to operating positions> or when the projector is repeatedly elevated 177 roils and depressed 408 roils or removed and reinserted into its orientation medium.

3.7.4 *Extraneous Light Emission - The light source shall not be visible except as observed through the lenses.*

3.7.5 Projected Reticle Image Definition - The projected reticle image shall be clearly defined with no visible breaks when viewed from a normal operating position against a bright background (such as the reflection of a clear sky from water) or when viewed in a darkened area.

3.7.6 *Projected Reticle Image Pattern Accuracy - The angle subtended by the innercircle of the projected reticle image pattern shall be 30 ± 1 roil.*

3.7.7 , Parallax - When viewing at a distance of eight to ten inches from the beamsplitter, parallax between the reticle image center and the target shall not exceed 2.0 roils over the full aperture of the beamsplitter when viewing a target at a range of 1000 ± 200 meters.

3.7.8 *Image Tilt - The vertical line of the projected reticle image shall be plumb within 1.0 degree when meeting 3.7.2.*

3.7.9 Electrical Controls - Rotation of the light intensity control, from stop to stop in a clockwise **(cw)** *and counterclockwise (CCW) direction, shall gradually vary reticle lamp light intensity in both positions of the filament selector switch. Full cw to full ccw rotation of this control shall produce a gradual and regular dimming of the reticle, with no breaks in continuity. with the control in an extreme* **ccw** position, filaments Of the dual filament reticle lamp shall emit no light.

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3.7.10 *Arm 10547473 Operability - When applied at the stop on Housing 10547482 casting, a torque of between 20 and 30 pound inches shall move Arm 10547473 0.002 inches from its operating position; a torque of between 45 and 60 pound inches shall move it from an operating to stowed position; and a torque of between 30 and 45 pound inches shall move it out of a stowed position.*

3.7.11 Sealing - Housing 10547482 shall evidence no leakage in excess of 0.2 pounds-per-square-inch, guage pressure (psig) when subjected to an internal pressure of 6 ± 1.0 psig for at least two hours.

3.7.12 Interior Atmosphere - The interior atmosphere of Housing 10547482 shall have a dew point of -25"F, or lower, with an atmospheric pressure of 0.10 to 0.25 psig.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-sTD-109.

4.2 *First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or* **contractor**) *for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) projectors plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials,* **equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.**

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements bf the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance~th the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 and MIL-O-13830 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with M1L-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval, compliance with certification requirements as specified in 4.4.3, and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. vibration B	**3.7.1, 3.7.1.2**	**4.6.4, 4.6.4.2**
102. Sealing	**3.7.11**	**4.6.5.1**
103. Interior atmosphere	**3.7.12**	**4.6.5.2**
104. Arm: 10547473 operability	**3.7.10**	**4.6.5.3**
105. Electrical controls	**3.7.9**	**4.6.5.4**
106. Extraneous light emission	**3,7.4**	**4.6.5.5**
107. Cleanliness	**3.3.l(b)**	**4.6.5.6**
108 . Optical-mechanical alignment	**3.7.2**	**4.6.6.1**
109. Projected reticle definition	**3.7.5**	**4.6.6.2**
110. Projected reticle accuracy	**3.7.6**	**4.6.6.3**
111. Parallax	**3.7.7**	**4.6.6.4**
112. Image tilt	**3.7.8**	**4.6.6.5**
113. Optical-mechanical alignment retention	**3.7.3**	**4.6.6.6**

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MINOR : NONE DEFINED)

NOTE: The tests in Table I shall be conducted in accordance with the test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

4.3.3 *Acceptance* and relection. Rejected lots shall be screened for all defective characteristics Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 *Special samPlin&.- Samples* **shall be selected from lots at random** *using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection* of the *lot* shall be in *accordance with AQL 2.5 of Table II-A of MIL-STD-105* The samples shall* **have met all** the requirements and tests outlined in Table I prior to and after being subjected to the following examinations and tests~

TABLE **II**

CHARACTERISTIC	*REQUIREMENT*	**TEST PROCEDURE**
301. **Storage at extreme temperature**	**3.4**	**4.6.1**
302. Operability at extreme temperature tures	**3.5, 3.5.1, 3.5.2**	**4.6.2, 4.6.2.1, 4.6.2.2**

4.4.1.1 Failure of samPle Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have *been made by the contractor and resubmitted samples have been approved (see 4.3.3).*

4.4.2 *First* **article samPle.- Unless otherwise specified in the con-** *tract, test(s) listed herein shall be conducted on* one "unit *of* **the** *initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.*

CHARACTERISTIC	**REQUIREMENT**	**TEST PROCEDURE**
303. *Humidity*	*3.6*	*4.6.3*
304. Vibration A	**3.7,1, 3.7.1.1**	**4.6.4, 4.6.4.1**

4.4.3 Certification.- Certification shall include test data and results for all characteristics specified in Table III. Certification shall be **provided** *prior to performing final acceptance inspection, and* re- *main* in effect throughout the contract for Government acceptance. This *does not reMove the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available* **to** *the Government. upon request. A new*

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certification will be required SHOULD any changes occur in materials, manufacturing processes and techniques previously certified.

TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Materials	3.1	Applicable drawings
Fabrication	3.2	Standards, and
General specification (MIL-F-13926)	3.3.2	Specifications
General specification (MIL-O-13830)	3.3.1 (except 3.3.lb)	

4.5 Inspection equipment.-- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements,- Inspection equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

a. Environmental and mechanical test equipment- Temperature, humidity, and vibration test equipment shall conform to applicable

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requirements in the "Test Facilities" requirement of MIL-F-13926C Standard vibration equipment provided shall have provisions for controlling the frequency and amplitude of vibration in accordance with 3.7.1.1 and 3.7.1.2. Vibration equipment shall have provisions for holding the projector securely throughout vibration tests, and allow its orientation in three mutually perpendicular planes for Vibration A and in one plane only for Vibration B to simulate the projector's position when mounted for use. A standard time measuring instrument shall be provided to control vibration periods. A multi-station pressure tester having at least the functional accuracies of Item 8 in 4.5.2.4 shall be provided along with a dew-point tester and regulator. A spring scale and gage (Item 4 in 4.5.2.4) to measure force values shall also be provided, Unless otherwise specified, test equipment herein shall be of standard commercial quality.

b. Optical test equipment.- Inspection equipment having at least the functional accuracies of Items 1, 2, and 3 in 4.5.2.4 shall be provided. A bench level and surface plate capable of being cross-leveled or other means to accomplish this objective, shall also be provided. On the surface plate, mediums shall be provided to hold, position, and secure, along specified geometric lines in compliance with 3.7, the projector-collimator, auxiliary telescope, and projector under test. Figure 1, the instructions herein or on test fixture drawings shall be utilized, as applicable, to accomplish the orientation in 3.7. When the crosshair intersection on retitles of an auxiliary telescope and 'rejector-collimator are in coincidence, the means provided in 4.5.2.4 shall assure that the line of sight obtained establishes line W in Figure 1 - in proper relationship to all other geometric lines depicted in Figure 1.

co Test targets.- The test target pattern shall contain two concentric circles, a vertical line and a horizontal line. Lines shall bisect each other at the center of all circles and line extremities shall exceed the larger circle's diameter. On a wall target, actual circle diameter shall equal the chord length of an arc which intersects both legs of the angle subtended at the target's location. This angle shall be 29 rolls for the smaller circle and 31 roils for the larger circle. The test target provided for the vibration test shall have a circle which subtends a 10 mil angle. Targets shall contain acceptance limits for optical requirements to be inspected. On wall targets, strategically located mil blocks may be provided and their size determined as follows:

$$\text{MIL BLOCK SIZE} = \frac{\text{TARGET DISTANCE (in inches)}}{1000}$$

d. Electrical test equiPment.- Ark electrical power source (Item 7, in 4.5.2.4) shall be provided to supply the direct current (de) potential in proper polarity required by individual tests. Electrical instrumentation (Item 5 and 6 in 4.5.2.4). shall be provided to measure and monitor the electrical performance of the projector under test, and flexible conductors shall be provided to connect electrical instrumentation

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installed clear of the vibration table. Electrical instrumentation shall be provided to verify the projector's electrical components and wiring that will not damage circuitry or components.

4.5.2.4 Special test equipment accuracy.- The special test equipment in 4.5.2.3 shall have at least the accuracy depicted for the following equipment:

ITEM	NOMENCLATURE	DESCRIPTION
1.	Inspection Aid (Gunner's Quadrant)	Conforming to Drawing A8658940
2.	Gage fixture optical	Depicted on Dwg. F10552O1O
3.	Gage Fixture optical	Depicted on Dwg. F10552O37
4.	Force gage	0.002 inch thick with attached weight of 2 to 4 grams
5.	Voltmeter	0-50 vdc; full scale accuracy ± 2%
6.	Ammeter	0-1.0 amp. dc; full scale accuracy ± 2%
7.	Electrical power supply	20 to 30 vdc adjustable output w/8Z regulation
8.	Pressure Tester - Two Station	Conforming to Drawing F8565556

4.6 Test methods and procedures.

4.6.1 Storage at extreme temPeratures- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the projector is thermally stabilized at each extreme temperature specified in 3.4. Maintain each extreme temperature for an uninterrupted period of at least 24 hours. Upon completion of each exposure the temperature shall be gradually returned to between +60°F and +90°F. Subsequent to completion of each temperature excursion, the projector shall be examined to determine that no evidence of damage exists.

4.6.2 Operability at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the projector is thermally stabilized at each extreme temperature specified in 3.5. Maintain each extreme temperature for only the uninterrupted period of time required to perform the tests and examinations specified herein. The tests and examinations in 4.6.2.1 to 4.6.2.2 shall be performed to verify manual operability of mechanical subassemblies at each extreme temperature in compliance with 3.5* Upon completion of all examinations at each extreme temperature, the temperature shall be gradually returned to between +60F and +90°F.

4.6.2.1 Light intensity control and light switch control.- Manually rotate the light intensity control knob over its entire range in a clockwise

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(CW) and counterclockwise (CCW) direction. Determine that no binding is evident in the control using normal hand pressure and that the light intensity control knob is secure on its shaft. The light switch toggle when manually operated shall be examined for ease of manipulation with positive indexing in both "ON" positions.

4.6.2.2 Arm 10547473.- Utilize the same test procedure specified in 4.6.5.3 and determine that Arm 10547473 torque measurements do not exceed the values measured in 4.6.5.3 by more than 10 pound inches at each extreme temperature in 3.5. Operating and stowed positions shall be as specified on Figure 1. Observe that Arm 10547473 moves freely between operating and stowed positions and determine that the mechanism provided will fasten it in both positions.

4.6.3 Humidity.- This test shall be performed subsequent to the sealing test and prior to the interior atmosphere test. Except as specified herein, conduct this test in accordance with the "Test Facilities" and "Humidity (Procedure 111)" requirements of MIL-F-13926. The projector shall be exposed for at least 24 hours to a 97% relative humidity (RH) at an ambient temperature of +160°F. Chamber temperature shall then be lowered to between+60° and +90°F with condensation and the projector exposed to this condition for at least four hours - within which time chamber temperature may be adjusted to obtain other condensation periods. The humidity phase of this test shall now cease and the external surface of the projector shall be dried by wiping all accessible areas - do not dry by air blast or oven. Protect electrical pins, not normally exposed, from humidity.

4.6.4 Vibration.- Prior to vibration, perform test 4.6.6.1, determine compliance with 3.7.2, and record the optical-mechanical alignment demonstrated by the projector. Vibration shall be under the test conditions in the "Test Facilities" requirement of MIL-F-13926, utilizing test equipment conforming to 4.5 and with 28 vdc applied to the projector's electrical input - in proper polarity. Throughout vibration, the projector shall be under constant surveillance to detect any evidence of optical, mechanical, or electrical component failure or malfunction. Except during dwells in 4.6.4.1, electrical circuits shall be monitored to verify that no input current change occurs - indicative of intermittent electrical connection, electrical short circuit, or component malfunction in either "ON" position of the filament selector switch. The constant current drawn by each filament of the dual filament lamp - or between lamps - need not be of identical value. Subsequent to vibration, the projector shall be examined for evidence of damage and again subjected to test 4.6.6.1 to determine compliance with 3.7.1.

4.6.4.1 Vibratian A.- Perform this test in compliance with requirements of 3.7.1 and 3.7.1.1 and in accordance with procedure I (Parts 1, 2 and 3) of MIL-STD-81O and the specific procedure herein and in 4.6.4. During this test, set the light intensity control for maximum reticle lamp illumination. The filament selector switch shall be positioned in an alternate "ON" position, each time Arm 10547473 is relocated during each first hour vibration in each plane. During the first hour of each two hour

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vibration in each plane, the arm (Arm 10547473) shall be in a stowed Posi tion for one half hour and in an operating position for the remaining half hour. Record throughout the first hour of vibration in each plane all predominant resonant frequencies exhibited by the projector and note the position of the arm at that time. Upon completion of the first hour of vibration in each plane, select the four most prominent resonant fre- quencies of all those recorded regardless of the position of the arm. Remove the reticle lamp and during the remaining one hour vibration in each plane, vibrate the projector for not less than 15 minutes at each one of the four resonant frequencies previously selected with the arm in the position noted when the resonant frequency was first recorded. Upon com- pletion of the entire vibration test (of two hours in each plane for a total of six hours), the projector shall be examined as specified in 4.6.6.6 with reticle lamp 10547031 reinserted.

4.6.4.2 Vibration B.- Perform this test in compliance with require- ments of 3.7.1 and 3.7.l.2 ,procedures herein and in 4.6.4 with the light in- tensity set for optimum viewing of the projected reticle image. Prior to vibration, the geometric center of the projected reticle image shall be made coincident to the intersection of a crosshair and this intersection shall be located at the center of a target's five mil radius circle. The filament selector switch shall be in one "ON" position for half the vibra- tion period and in an alternate ON position for the remaining half. Throughout vibration, the observer shall locate his eye eight to ten inches from the beamsplitter in the direction of the projector lamp housing to verify that no flicker occurs in the projected reticle image and that the coincidence established herein remains within the circumference of the ten mil target circle. Upon completion of the entire Vibration test, the subsequent to vibration examination specified in 4.6.6.6 shall be per- formed.

4.6.5 General.

4.6.5.1 Sealing.- This test shall be performed prior to the humidity test 4.6.3 - when a humidity test is required (see 4-4) utilizing test equipment conforming to 4.5. Housing 10547482 shall be internally pressurized with dry nitrogen gas to the pounds-per-square-inch gage pressure (psig) in 3.7.11 for not less than two hours. Throughout this period, no pressure drop greater than the limit specified in 3.7.11 shall be evident. Upon completion of this test, flush the housing with dry nitrogen gas through a regulator to prevent excessive internal pressure during flushing. Utilize a dew point tester and continue flushing until the gas bled from the housing exhibits a dew point at least as low as -25°F with an internal pressure of between 0.10 and 0.25 psig. Housing 10547482 shall then be sealed.

4.6.5.2 Interior atmosPhere.- This test shall utilize the procedure in 4.6.5.1 except that the examination previously performed to detect ex- cessive loss of pressure need not be repeated. this examination shall verify compliance with 3. 7.12 and be performed subsequent to all tests and examinations of 4.6 except 4.6.5.6.

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4.6.5.3 Arm 10547473 oPerabilitY.- This test consists of applying normal to Arm 10547473 the direct forces that equal the torque values in 3.7.I.O. Actual values of direct force required for each examination shall be determined by the formula:

$$\text{DIRECT FORCE APPLIED (pounds)} = \frac{\text{TORQUE VALUE IN 3.7. 10 Pound inches)}}{\text{ARM LENGTH UPON WHICH FORCE ACTS (inches)}}$$

Arm length upon which force acts shall be measured along Arm 10547473 toward the beamsplitter from an axis (line D in Figure 1) through the center of the bore in the yoke casting supporting Arm 10547473. Place Arm 10547473 in an operating position with the test gage (see 4.5.2.4) inserted - weighted end down " between the stop surfaces. Use a spring scale (see 4.5.2.4) and apply sufficient force to cause the shim to fall clear. The measured value of slowly applied force that moves Arm 10547473 a distance in excess of 0.002 inch, which allows the shim to fall clear, shall be used to verify compliance with the applicable requirement in 3.7.10. Now use the spring scale and measure the force required to freely move Arm 10547473 from (not out of) its operating position to its stowed position and verify compliance with the applicable requirement in 3.7.10. Arm 10547473 shall then be placed in a stowed position. Again use the spring scale and measure the force required to cause the restraining linkage to fold permitting Arm 10547473 to move freely toward its operating position.

4.6.5.4 Electrical controls.- Apply 28 vdc to the projector, position filament selector switch sequentially in each "ON" position and rotate the light intensity control knob from its extreme ccw position of minimum lamp illumination to the extreme cw position of maximum lamp Illumination. Determine that a progressive increase in lamp emitted light occurs with cw rotation of the control and a progressive decrease with ccw rotation - in both positions of the filament selector switch. With the control in an extreme ccw position and the switch sequentially placed in alternate "ON" positions, it shall be observed in a darkened area (having not more than 0.01 foot candle of illumination) that no light is emitted from the lamp. Obseved compatibility between control performance and markings and that the knob is secure to its shaft. Observe positive switch indexing and ease of manipulation of both controls.

4.6.5.5 Extraneous light emission.- Energize the projector with 28 vdc and adjust the light intensity control to obtain maximum light output from the reticle lamp in a darkened area (as specified in 4.6.5.4). Perform a visual examination of the projector to determine that no light is emitted from the projector except through the lenses as defined by the limits prescribed by the reticle drawing (D10543371).

4.6.5.6 Cleanliness.- Subsequent to conducting all tests and examinations, optics of the projector shall be examined in accordance with MIL-G 13830 to verify compliance with 3.3.lb.

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4.6.6 Optical Utilize test equipment conforming to 4.5, test procedures herein, and test set-up instructions In subparagraphs (a) and (b) or on test fixture drawings, as appropriate, to perform the following tests.

a. Holding medium set-up instructions.- On a cross-leveled surface mount the holding mediums provided for the projector-collimator, auxiliary telescope and projector. Locate and identify with appropriate temporary marks the axes of the projector which corresponds to lines A, C, and D (Fig. I). Since the centerline (/of datum diameter C represents the geometric longitudinal axis of the main housing 10547482 (Drawing F10547482) and is not readily identified or marked, the surface on the main housing designated as surface X on this drawing may be used to locate the @ of datum diameter C. With appropriate marks, establish on the holding mediums two horizontal planes 4.5-inches apart, that are parallel to each other and parallel to the cross-leveled surface plate. The upper plane shall contain line W and the lower plane line A 90° to line W. Utilize the 0.250 inch diameter hole in the main housing 10547482 and on the projector's holding medium, establish line D in a plumb vertical transverse plane 90° with respect to longitudinal line W and the surface plate. Now establish the diagonal longitudinal plumb vertical plane containing line C which shall form an angle of 40° ± 2° with respect to a plumb vertical longitudinal plane containing line D. This plumb vertical longitudinal plane containing line D shall then be employed to establish a second plumb vertical longitudinal plane parallel to the former at a distance of 17-5/16 inches which shall also contain line W. Make the appropriate marks on the mediums provided to hold the auxiliary telescope and projector-collimator to facilitate proper orientation of both instruments with respect to the line W as established herein.

b. Instrument set-up instructions.- Prior to inserting the projector into its holding medium, mount the auxiliary telescope(s) and projector-collimator and verify that their retitles are aligned and superimposed on each other and orientate the geometric axis (line W in Figure 1), so obtained, in respect to other axes to provide the line of sight required to verify compliance with 3.7.2 and 3.7.3* The projector shall now be mounted in its holding medium with Arm 10547473 in an operating position. The projector shall be energized through its electrical input connector with 28 vdc in proper polarity.

4,6.6.1 Optical-mechanical alignment.- Adjust the light intensity control for optimum viewing and sight along the geometric axis (Line W in Figure 1) established in 4.6.6. Examine the coincidence existing between the center of the projector's projected reticle image, at the intersection of the crosshairs, and the" established line of sight to verify compliance with 3.7.2. This test shall be performed and the results obtained recorded prior to the vibration tests in 4.6.4 and the optical-mechanical alignment retention tests in 4.6.6.6.

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4.6.6.2 Projected reticle Pattern definition.- In a darkened area - as specified in 4.6.5.4 - adjust the light intensity control to obtain optimum viewing of the projected reticle pattern image, and conduct an examination to determine that its pattern is without breaks and clearly defined. After a lapse of 15 minutes - a light adaption period - employ a uniformly illuminated field of view of at least 2300 foot lamberts. The light intensity control shall now be adjusted to determine that a position exists where the projected reticle-pattern image is clearly visible.

4.6.6.3 Projected reticle pattern accuracy.- Adjust the light intensity control for optimum viewing and establish coincidence between the intersection of the horizontal and vertical lines on the target and those of the projector's reticle image. Examine the entire circumference of the projector's 30 mil circle to determine that it is within the 29 mil end 31 mil concentric target circles.

4.6.6.4 Parallax.- The observer shall locate his eye eight to ten inches from the beamsplitter at the Line W axis established in 4.6.6 and adjust the light intensity control for optimum viewing. Align both the vertical and horizontal reticle line of the projected reticle image close to but not touching the vertical and horizontal lines on the targets - located at an actual or simulated distance of both 500 and 1000 meters. The observer shall slowly move his head up and down and from right to left - observing both target lines and both reticle lines. The observer shall 1 verify by this examination that no line displacement occurs in excess of the limit specified over the full aperture of the beamsplitter when viewing actual or simulated 1000 ± 200 meter target.

4.6.6.5 Image tilt.- Perform this test with a target line that is plumb, aligned' with the vertical longitudinal plane containing the Line W axis established in 4.6.6, and with one extremity of the projector's reticle image vertical line in coincidence with the target's plumb line - at the bottom of the 31 mil circle. The other extremity of the projector's reticle image vertical line shall now be observed at the top of the 31 mil circle to verify that it is not displaced more than 0.4 mil from the plumb target line.

4.6.6.6 Optical-mechanical alignment retention.- This test consists of repeating 4.6.6.1: (a) subsequent to vibration tests in 4.6.4, (b) UPOn elevating and depressing the projector (in accordance with the procedure specified herein) and (c) after completion of each one of each three cycles of the handling cycles specified herein. The recorded data in 4.6.6.1 shall be utilized to verify compliance with 3.7.1 and 3.7.3.Elevation and depression of the projector shall consist of rotating the projector's axis (represented by line D in Figure 1) - in a vertical longitudinal plane 177 mils from a plumb line toward the observer and then 408 roils from a plumb line away from the observer about an axis which is 90° to line D, parallel to line A (Figure 1), and in the same horizontal longitudinal plane as A. This rotation will elevate and depress the beamsplitter, during

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which time vertify no displacement (other than rotation herein) in axis A from its initial position. Prior to and after completing each operational cycle, as defined herein, verify that a line of sight along the established geometric axis (line W in Figure 1) intersects the center of the projected reticle image in elevation and depression within specified limits. Each handling cycle shall consist of moving Arm 10547473 from its operating to stowed to operating positions; elevating and depressing the projector as previously specified in this test; and removing the projector from its medium and again reinserting and repositioning the projector in its holding medium.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing, and Marking.- Packaging, packing and marking of Projector, Reticle Image: 10547353 shall be in accordance with packaging data sheet **10547353.** The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use - The projector covered by this specification is intended for use with a support and bracket assembly as an integral part of Sight, Reflex: XM70E1 (MIL-S-60828), which is used by the pilots of (OH-6A and OH-58A) helicopters to obtain a high probability first round hit.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable packaging data sheet number (see 5.1)
d. Applicable stock number
e. Requirement for First Article Testing (see 4.2)

Custodian:

Army-MU

Preparing activity:

Army -MU

Project No. 1270-A034

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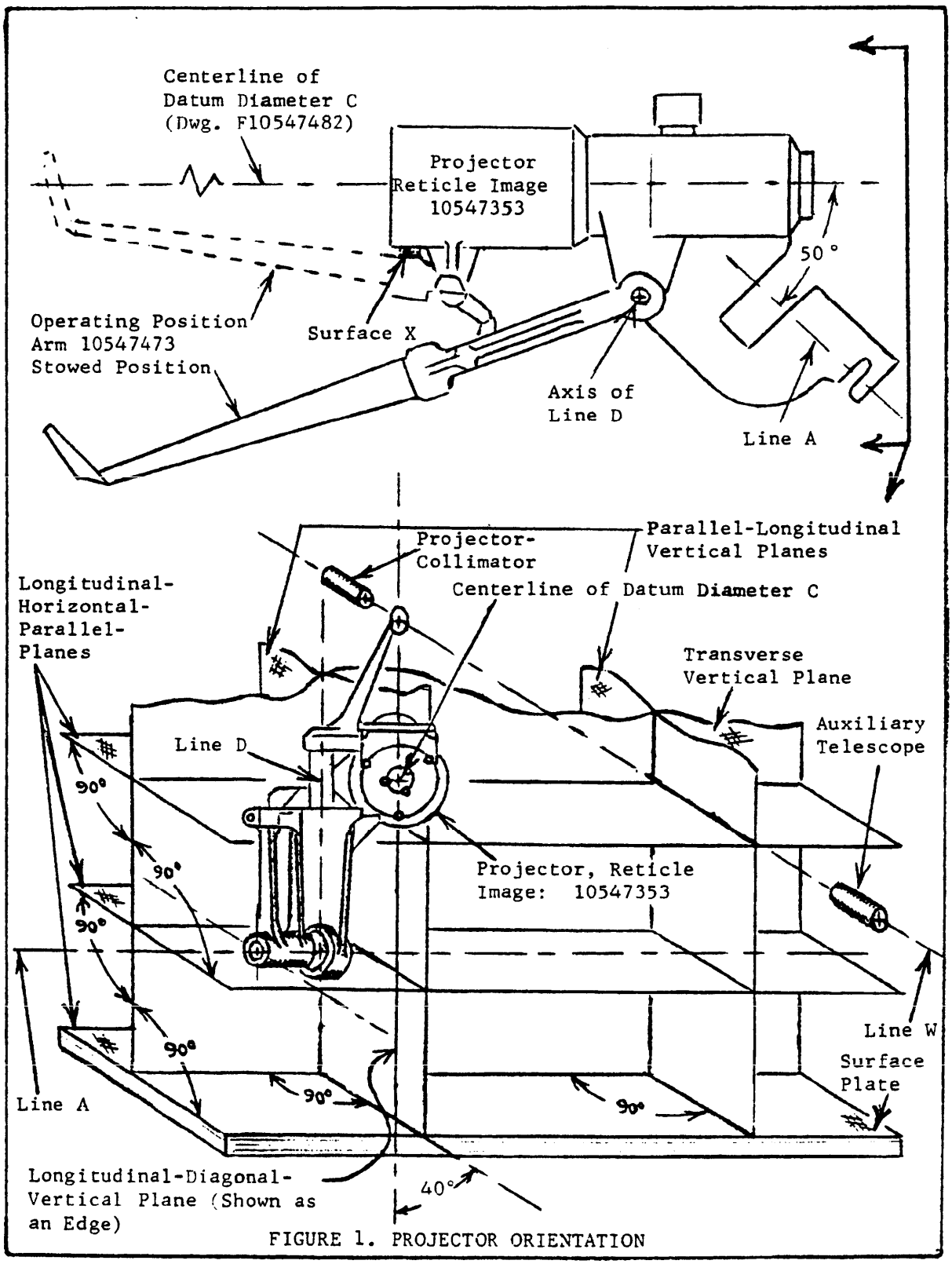


FIGURE 1. PROJECTOR ORIENTATION

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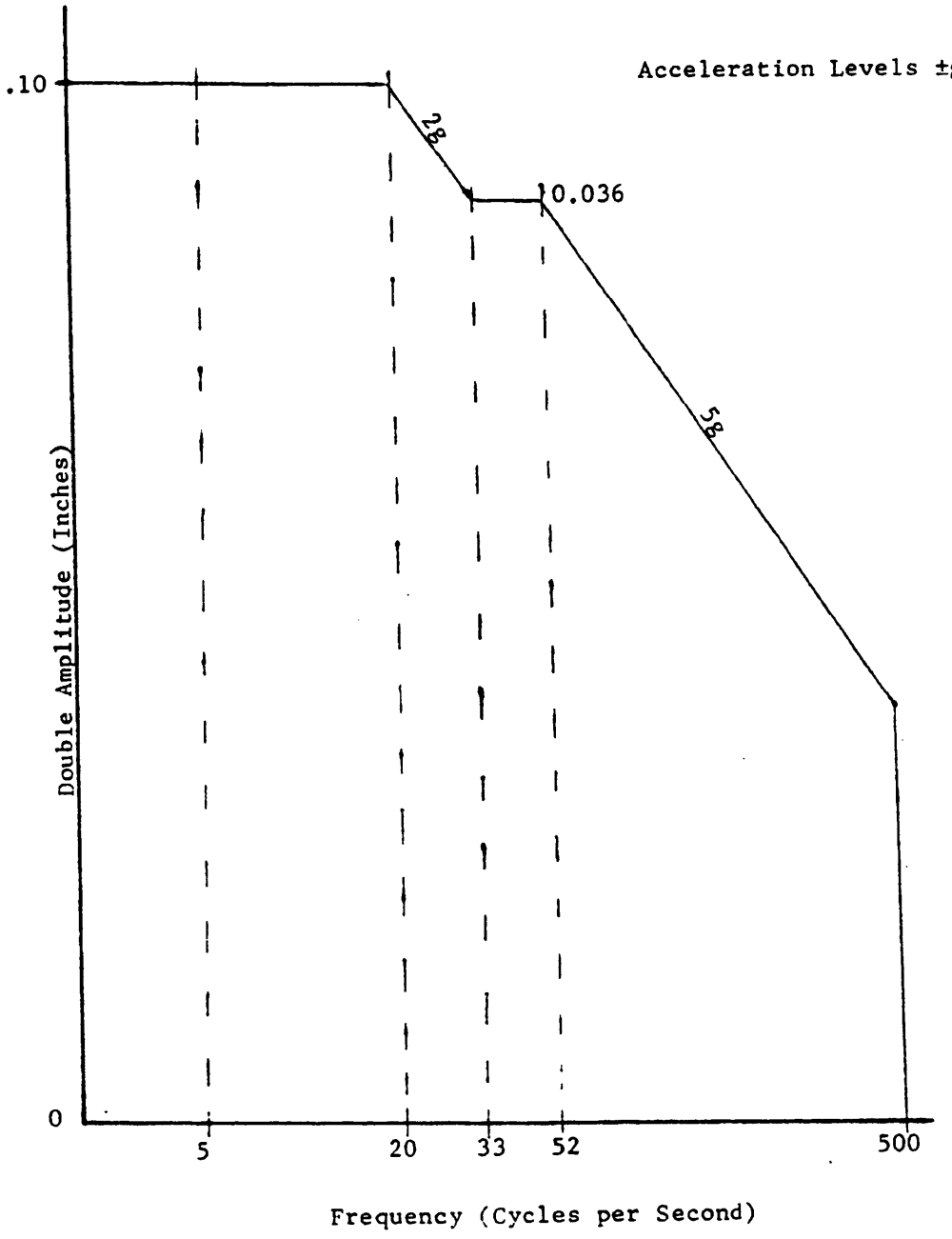


FIGURE 2. VIBRATION A CURVE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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